

Measuring the Dispersion of Data: Quartiles & Boxplots

- ❑ **Quartiles:** Q_1 (25th percentile), Q_3 (75th percentile)
- ❑ **Inter-quartile range:** $IQR = Q_3 - Q_1$
- ❑ **Five number summary:** min, Q_1 , median, Q_3 , max
- ❑ **Boxplot:** Data is represented with a box
 - ❑ Q_1 , Q_3 , IQR: The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
 - ❑ Median (Q_2) is marked by a line within the box
 - ❑ Whiskers: two lines outside the box extended to Minimum and Maximum
 - ❑ Outliers: points beyond a specified outlier threshold,
plotted individually
 - ❑ **Outlier:** usually, a value higher/lower than $1.5 \times IQR$

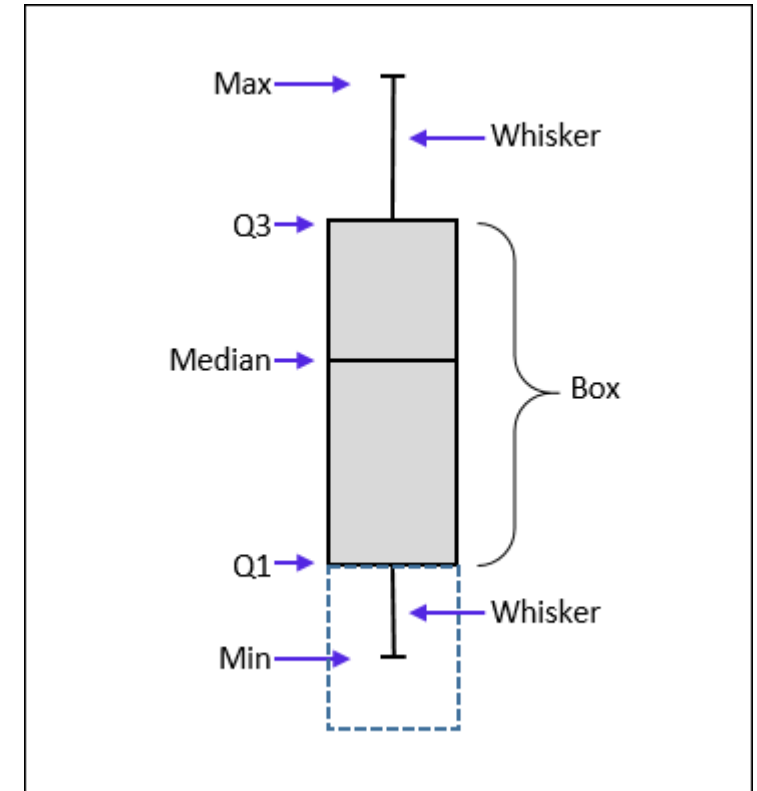


Image Source: <http://blog.contextures.com/wp-content/uploads/2013/06/boxplotsimple08.png>